

Jehyun Park

Github: github.com/Parkprogrammer / **Blog:** parkprogrammer.github.io/

E-mail: jaheon555@g.skku.edu / **Linkedin:** <http://www.linkedin.com/in/je-hyun-park>

RESEARCH INTERESTS

Theoretical foundations of neural network generalization and optimization
Q-Learning based Optimization of Reinforcement Learning
Physical Simulation of Dynamics and AI Systems

EDUCATION

Sep 2026 — Present	<i>Seoul National University</i> <i>M.S., Interdisciplinary Program in Artificial Intelligence</i> <i>Advisor: Professor Youngjae Yu</i>	<i>Seoul</i>
Mar 2020 — Feb 2026	<i>Sungkyunkwan University</i> <i>BS., Computer Science and Engineering</i>	<i>Suwon, Korea</i>

EXPERIENCES

Dec 2025 – Sep 2026	Research Intern, Seoul National University SNU-PI Lab - Conducting Research in Optimization, Reasoning - Optimizing Sequence-Modeling architectures (Transformer, SSMs) and Optimizers - Multi-modal reasoning with Representation Learning and Reinforcement Learning - Under the supervision of Prof. <i>Youngjae Yu</i>	<i>Seoul</i>
Mar 2025 — Aug 2025	Undergraduate Research Intern, Yonsei University ML/AI Lab - Conducted Research in Few-shot Learning, Task Vectors, Test-Time-Training - Implemented algorithms and methods on task vectors in In-Context-Learning - Focused on Solving Out-Of-Distribution Problem, Catastrophic forgetting - Under the supervision of Prof. <i>Kyung Woo Song</i>	<i>Seoul</i>
Sep 2024 — Feb 2025	AI Researcher, The Catholic University of Korea St. Vincent's Hospital - Implemented machine learning algorithms for fall risk prediction protocol - Handled imbalanced patient data with K-NN imputer, t-SNE, Sent2Vec embeddings. - Implemented Imbalanced-XGBoost based pipeline and achieved the best <i>f1-score</i>.	<i>Suwon</i>
Jun 2023 — Aug 2023	Industry-Academia Collaboration Software Engineer, SOYNET - Participated in a year-long Industry-Academia Collaboration Project between SKKU & <u>Soynet</u> - Worked at the Company as a Software Engineer for 2 months - Ported Pytorch-based models to Tensorflow-based SOYNET Acceleration Engine - KIST University Person RE-ID model Optimization Applied at National Safety SQI Soft Task	<i>Seong Nam</i>

AWARDS & PROJECTS

Jan 2026	2025 KSC Excellence Award, Undergraduate Division	<i>Yeosu</i>
	• Excellence Award in Undergraduate Paper Competition • KSC 2025 held at Korean Institute of Information Scientists and Engineers • Received for paper “<i>Adaptation in Out-Of-Distribution via Sparse Autoencoders in Vision Transformers</i>”, sole author in Undergraduate division.	
Aug 2025 — Sep 2025	2025 Samsung AI Challenge 1st Prize Winner	<i>Suwon</i>
	• Theme: Methods to Reduce Large Model Size Without Performance Degradation • Individually competed on optimizing and compressing Large Language Model • Implemented <i>Context-aware-Expert-Merging-and-Pruning</i> Method on Qwen-30B-A3B Model • Compressed 30B Parameters to 23.9B by training-free Expert Merging and Pruning • 1st prize winner and Awarded 7,500\$ from Samsung AI Center	
Jul 2024 — Feb 2025	20th Korean Economic Securities Derivatives Competition 2nd Place	<i>Seoul</i>
	• Designed an ETF for IPO stocks in KOSPI & KOSDAQ Stock Market as part of a Team • Used GMM and t-SNE, Weighted K-NN for for Unsupervised similarity analysis • Portfolio Revenue Optimization using Disciplined Convex Programming and SLSQ Programming • 2nd place and Awarded 4,500\$ with Korea Exchange Chairman's Award	
Jul 2024 — Jan 2025	LLM-based Recommendation System Health care Startup, iKooB	<i>Seoul</i>
	• “Exploration of AI model architecture based on transfer learning Development of model data management platform” Collaborating with <u>iKooB</u> • Optimized a finetuned Llama-3 based 8B model on medical domains, implemented TF-IDF method and Chain-of-Thought (COT), Prompt tuning for Prototype model • Received Engineering Innovation Award from SungKyunKwan University	
Dec 2023 — Jan 2024	Winter Global Capstone Design Project, AIIT & SKKU	<i>Tokyo, Japan</i>
	• Collaborated with Tokyo AIIT University and participation in seminars and presentations • Presented an effective construction scenario for signaling system cloud infrastructure for mobility Scenario proposal for integrated cloud and AI-based solutions	
Sep 2023 — Dec 2023	2023 NH Investment & Securities Big Data Competition <i>Finals</i>	<i>Seoul</i>
	• Designed and trained a Vanilla Transformer Encoder for correlation analysis (<u>Fin2Vec</u>) • Used 1D-CNN, GRU for stock price data compression and tokenization • Implemented Self-Supervised Learning algorithm based on Self-Distillation and EMA (data2vec)	
Jun 2023 — Dec 2023	Development of SNS platform exclusively for college students	<i>SKKU</i>
	• Sungkyunkwan University Startup Support Group: A SNS App development project: (<u>CLIPPED</u>) • Participated as a Project Manager and Frontend Developer • Implementation stack: Flutter, AWS, next.js	
Jan, Sep 2020 / Sep 2023	<i>Sungkyun Software Scholarship</i> for bachelor's scholarship	<i>SKKU</i>
	• <i>Sungkyun Software Scholarship</i> Received from SungKyunKwan University • 3000\$ for each 3 Semesters, total amount of 9000\$ scholarship received.	

PUBLICATIONS

Adaptation in Out-Of-Distribution via Sparse Autoencoders in Vision Transformers

Korean Institute of Information Scientists and Engineers 2025 (bachelor's thesis')

Je Hyun Park*

Development of an Artificial Intelligence Algorithm for Fall Prediction (Poster Presentation)

Korean Society of Medical Informatics (KOSMI) Fall Conference, November 2025

Jaheon Park*, Yeon Woong Chung, M.D., Ph.D.^{†‡}, Seung Hyun Ko, M.D., Ph.D. Ph.D.[§]

Development of an Artificial Intelligence Algorithm for Fall Prediction (Poster Presentation)

Korean Society of Artificial Intelligence in Medicine (KOSAIM) Annual Conference, October 2025

Yeon Woong Chung, M.D., Ph.D.^{†‡}, **Jaheon Park***, Seung Hyun Ko, M.D., Ph.D. Ph.D.[§]

SKILL	Programming		
	Python	Various machine learning & processing libraries such as <u>Pytorch</u> , <u>scikit-learn</u> , <u>Numpy</u> , cupy, cxyppy	3
	C, C++	Conducted many system-architecture college projects XV6, Spike-simulator, ns3, mini-shell	2
	CUDA	Optimization of pytorch functions : im2col, and XGBoost Minimal Variance Sampling methods	1
	Next, Node.js	Web Projects using Next.js and Node.js with tailwind-css	1
	Flutter	Developed front-end app for the SNS platform	1
	Language		
	Korean (Native)		
	English (Fluent) (TOEIC: 930)		
